

Embolization of the Middle Meningeal Artery for the Prevention of
Chronic Subdural Hematoma Recurrence in High Risk Patients
« EMPROTECT »

PROTOCOL FOR INTERVENTIONAL RESEARCH INVOLVING THE HUMAN
PERSON ON A MEDICAL DEVICE OR ON AN IN VITRO DIAGNOSTIC MEDICAL
DEVICE

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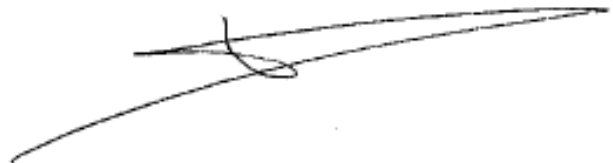
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The research will be carried out in accordance with the protocol, current good practice and the legislative and regulatory provisions in force.

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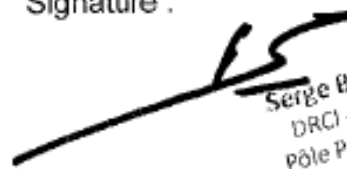
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1 SYNOPTIC SUMMARY

Full title	<i>Embolization of the Middle Meningeal Artery for the Prevention of Chronic Subdural Hematoma Recurrence in High Risk Patients</i>
Acronym	EMPROTECT
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Sponsor	<i>Assistance Publique - Hôpitaux de Paris</i>
Scientific justification	<i>Chronic subdural hematoma (cSDH) is the most frequently encountered pathology in emergency neurosurgery. The standard treatment is burr-hole evacuation, with placement of a temporary drain in the subdural space. Post-operative recurrence, estimated at between 10% and 20%, is a major management challenge, due to its frequency and the associated morbidity and mortality. Embolization of the middle meningeal artery (MMA) has recently been proposed as a treatment for cSDH. We hypothesize that embolization of the MMA homolateral to a surgically evacuated cSDH in patients at high risk of recurrence would significantly reduce the risk of postoperative recurrence.</i>
Objective and primary endpoint	The main objective of the project is to evaluate the efficacy of MMA embolization in reducing the risk of 6-month post-operative recurrence of evacuated cSDH in patients at high risk of recurrence compared with usual management. The primary endpoint is the rate of cSDH recurrence 6 months after index burr-hole surgery, defined as: - the reappearance of a homolateral cSDH with a midline shift ≥ 5mm or a symptomatic homolateral cSDH, including leading to death - or, the presence of a homolateral SDH > 10mm in maximal thickness on the 6 months control head CT scan - or, the need for repeated surgery for a homolateral cSDH recurrence - or, the need for a new hospital admission in relation to a homolateral cSDH recurrence
Secondary objectives and endpoints	The secondary objectives are to evaluate the efficacy of MMA embolization in patients at high risk of recurrence compared with usual management on the following criteria: - the rate of repeated surgery for a homolateral cSDH recurrence during the 6 months follow-up period - the rate of disability and dependency at 1 and 6 months, defined by a modified Rankin Scale (mRS) score ≥ 4.

	- mortality rate at 1 and 6 months - the total cumulative duration of hospital stay, during the 6 months follow-up period, directly or indirectly related to the cSDH We will also evaluate the security of the procedure (occurrence of minor and major embolization procedure-related complications).
Experimental design	Multicenter, open-label, randomized controlled trial with two parallel arms
Population concerned	cSDH patients at high risk of postoperative recurrence
Inclusion criteria	Patients : - aged ≥ 18 years, - operated for a cSDH recurrence or operated for a first episode of SDH at high risk of recurrence (Chronic alcoholism, liver cirrhosis, effective anticoagulant therapy, antiplatelet therapy, thrombocytopenia or surgery without use of external drain). - with affiliation to a social security scheme ((beneficiary or beneficiary's beneficiary) (excluding Aide Médicale d'État) - having signed an informed consent form or for whom a delegate close relative or a support person has signed an informed consent
Non-inclusion criteria	Patients with: - cSDH evacuation by craniotomy or twist-drill craniostomy rather than burr-hole surgery - beyond 7 days after the index surgery (initial surgery or surgery for a recurrent cSDH) - functionally dependant patient before the cSDH with an mRS score ≥ 4 - life expectancy < 6 months - renal failure as defined by a creatinine clearance < 30 ml/min - pregnancy - history of allergy to a iodinated contrast agent - contraindication to both conscious sedation and local anesthesia (ex: agitation making the procedure impossible under local anesthesia or conscious sedation) - refusal to participate - patient for whom follow-up is deemed problematic (living abroad or homeless for instance) - under legal guardianship or trusteeship
Procedure	Embolization of MMA with calibrated particles under local anesthesia or conscious sedation within 7 days of surgical evacuation of the cSDH.
Comparator group	Standard care
Other acts added by search	Pre-embolization CT angiography of the supra-aortic trunks and circle of Willis in the intervention arm.
Expected benefits for participants and society	A demonstration of the efficacy of MMA embolization with calibrated particles would have a major impact on the management of chronic subdural hematomas (cSDH), with a reduction in the risk of post-operative recurrence, which is currently around 10-20%. Each recurrence is associated with a significant risk of mortality, disability and additional dependency. Our intervention, if effective, would

	<i>reduce the mortality and sequelae associated with these recurrences. It would also reduce the rate of re-intervention and length of hospital stay. The procedure would also allow the resumption of anti-coagulants and antiplatelet agents for primary or secondary prevention in cases of formal indication.</i>
Risks added by research	<i>This minimally invasive procedure would expose the patient to low risk (risk associated with endovascular navigation, exposure to X-rays and injection of a contrast medium). The procedure would require strict bed rest for 12 to 24 hours, and should not prolong hospitalization.</i>
Practical considerations	<i>In each participating center, there will be a pair of investigators consisting of an interventional neuroradiologist (INR) and a neurosurgeon. After evacuation of the cSDH by burr-hole or trephine, eligible patients will be identified by the neurosurgeon, who will contact the corresponding INR. The INR will provide clear information to the patient and/or family concerning the risk of recurrence of the cSDH, the possibility of performing a MMA embolization, the modalities and risks of this procedure, and the possibility of participating in the EMPROTECT trial. After signing an informed consent form (by the patient or a family member/trusted support person), eligible patients will be randomized 1:1 between embolization (experimental group) and usual care (control group). Randomization will be carried out in random permuted blocks, stratified by center, anticoagulant/antiplatelet treatment and whether the cSDH is unilateral or bilateral. It will be centralized via the e-crf. Patients randomized to the experimental group will undergo embolization under local anesthesia or conscious sedation, after CT angiography, within 7 days of surgical evacuation of the cSDH with strict bed rest for 12 to 24 hours afterwards. Patients in the control group will receive usual care. Patients will be reassessed at 1 and 6 months, and data will be collected via the e-crf. Blind validation of recurrences will be ensured by an independent adjudication committee.</i>
Number of subjects selected	342
Number of centers planned	12 centers
Duration of research	<i>Specify :</i> - inclusion period: 24 months - duration of participation (treatment + follow-up): 6 months - total duration: 30 months
Number of planned inclusions per center per month	1.2 on average
Statistical analysis	<i>To demonstrate a reduction in the recurrence rate from 15% (control group) to 5% (embolized group) with 80% power, and ensuring an overall bilateral alpha risk of 5%, by taking into account 2 sequential tests planned according to the Lan and Demets method with an O'Brien & Fleming-type alpha risk expenditure function (interim analysis after inclusion and end of follow-up of 129 patients (information fraction of 37.5%) at the nominal threshold $\alpha=0.001$</i>

	<i>and final analysis at threshold $\alpha=0.05$) and a lost to follow-up rate of 20%, the required number of patients is 342.</i> <i>The primary endpoint will be analyzed in the intention-to-treat population using a mixed logistic regression model adjusted for stratification factors (uni/bilateral nature of cSDH and anticoagulant/anti-aggregant treatment) with fixed effects and including a random effect on center.</i>
Source of funding	<i>Interregional PHRC</i>
Data Safety Monitoring Board	Yes

2 SCIENTIFIC JUSTIFICATION OF RESEARCH

2.1 Description of knowledge relating to the pathology concerned

Chronic subdural hematoma (cSDH) is a collection of aged (liquefied) blood between the dura mater and arachnoid. It usually occurs two to four weeks after minor head trauma. Radiological diagnosis is straightforward, and relies on a non-contrast computed tomography (CT) scan showing an extra-axial collection of variable density, usually unilateral and crescent-shaped along the convexity. This may be associated with deviation of midline shift, or even brain herniation. The most frequently encountered factors favoring the occurrence of CSDH are chronic alcohol abuse, advanced age, cerebral atrophy, and treatment with anticoagulants or antiplatelet agents (1–3). Following the formation of an acute subdural hematoma, lysis of the clot releases fibrin degradation products, triggering a local inflammatory process. This inflammation promotes neovascularization of the outer membrane of the cSDH in contact with the dura mater, itself richly vascularized (4–6). Microbleeds from this outer membrane are responsible for the secondary maintenance of the cSDH.

CSDH is one of the most frequently encountered pathologies in emergency neurosurgery. The reported annual incidence varies from 5 to 14 per 100,000 people (7,8). This range is at least partly linked to differences in demographic structure between populations, since incidence rises sharply above the age of 65, to 58 per 100,000 people per year (8). The increased incidence of cSDH is also linked to the use of anticoagulants and antiplatelet medication in this population. (2). An increase in cSDH incidence is therefore expected due to the ageing of the population and the growing use of anticoagulants and antiplatelet medication, particularly in the French population. This makes the management of cSDH a major public health issue. CSDH is associated with a mortality rate of 4% and a morbidity rate of 10%. (9). The standard treatment for cSDH is neurosurgical burr-hole evacuation with a drain. The risk of post-operative recurrence is around 10-20%, occurring mostly within 3 to 6 months (9,10). Known risk factors for recurrence include: 1) previously recurrent cSDH, 2) anticoagulation, 3) antiplatelet medication, 4) chronic alcohol abuse and 5) pre- and post-operative volume of the cSDH (11–14). Recurrence is itself a major risk factor for in-hospital morbidity and mortality. Preventing recurrence is therefore a major issue in practice and in clinical research.

2.2 Research hypotheses

The pathophysiology of cSDH involves inflammation with hypervascularization of the dura mater and hemorrhagic suffusion into the subdural space (15). The dura mater of the convexity is vascularized preferentially by the middle meningeal artery (MMA), whose caliber is generally increased on the side of the cSDH (16). Endovascular embolization of this artery with calibrated particles has recently been described for the treatment of cSDH, to prevent recurrence in simple or refractory cSDH, or even as a curative treatment for cSDH without threatening mass effect for some authors (15,17). Most of these publications report single cases or case series. A large series of 72 non-randomized cases was reported by Ban et al. (15). This team proposed embolization to all patients managed for cSDH, whether operated or not. Preliminary results are very promising. The procedure is straightforward and carries a very low risk of complication.

We hypothesize that MMA embolization of a surgically evacuated cSDH, in patients at high risk of recurrence, would significantly reduce the risk of postoperative recurrence.

2.3 Summary of relevant pre-clinical experiments and clinical trials

Improving the treatment of cSDH is an active area of research. For example, temporary placement of a drain in the subdural space after cSDH evacuation, compared with evacuation alone, has been shown to reduce the risk of recurrence, making it the standard surgical technique (9,18,19). No other medical or surgical intervention has yet proved effective in reducing the risk of recurrence. Recurrence can occur even after drain placement, particularly in patients with risk factors such as anticoagulant or antiplatelet therapy.

MMA embolization, most often with calibrated particles, but also with coils or cyanoacrylic glue, has been described for the treatment of cSDH to prevent recurrence (12,20–22). Most of these publications concern single cases or case series. This technique was initially proposed for very limited indications, namely refractory cSDH (after several recurrences), or after a first recurrence in patients with additional risk factors (12,17,23). Ban et al. extended the indications for embolization by proposing this technique to a consecutive series of patients with operated and unoperated (curative intervention) cSDH of more than 10mm thickness (15). Comparison of this cohort with a historical series suggests a significant reduction in recurrences and recourse to surgery (15). Finally, a small non-comparative study, referenced on *ClinicalTrials.gov* (<https://clinicaltrials.gov/ct2/show/NCT03307395?cond=subdural&rank=1>), is currently

evaluating embolization in the curative treatment of cSDH. To our knowledge, no randomized trial has yet been conducted to evaluate this procedure.

2.4 Description of the population to be studied and justification for its choice

The most common factors favoring the occurrence of cSDH are chronic alcohol abuse, advanced age, cerebral atrophy, treatment with anticoagulants or antiplatelet medication (1–3). The majority of patients will be relatively elderly. Many of these are vulnerable patients for whom any new surgery, hospitalization or neurological event can have a major impact on their life trajectory, leading to loss of autonomy, institutionalization, cascading complications or even death.

We propose to include patients aged 18 or over at risk of recurrence, i.e. those who have undergone surgery for a recurrence of cSDH or after a first cSDH surgery in the case of favoring factors (chronic ethylism, hepatic cirrhosis, effective anticoagulation or antiplatelet treatment, thrombocytopenia or cSDH surgery without drain placement). An intervention to reduce postoperative recurrence in this population would have a major impact. Approximately one-third of patients undergoing surgical management for cSDH have at least one of these characteristics.

Patients operated by craniotomy will be excluded. Craniotomy is not the procedure of first choice for cSDH surgery due to the higher morbidity associated with this more invasive surgical procedure, although the risk of recurrence is lower compared with burr-hole evacuation according to the meta-analysis by Almenawer et al. (9). Craniotomy with membranectomy in the treatment of cSDH is therefore generally reserved for rare indications, such as refractory organized cSDH as described by Matsumoto et al. (17).

We will also exclude patients who are dependent before the cSDH, with an estimated life expectancy less than 6 months, with chronic renal failure contraindicating the injection of iodinated contrast media (clearance of less than 30 ml/min or allergy to a contrast medium), with an active pregnancy contraindicating exposure to x-rays, agitated (making it impossible to perform the procedure under local anaesthesia or conscious sedation), refusing inclusion or for whom follow-up is impossible.

2.5 Description of the embolization procedure

Patients randomized to the experimental group will have embolization under local anesthesia or conscious sedation, after CT angiography of the supra-aortic trunks and circle of Willis, within 7 days of surgical evacuation of the cSDH. The CT angiography will enable us to verify the feasibility of arterial catheterization and embolization. If this is not the case, embolization will not be performed. In particular, the CT angiography will be used to check for the presence of a floating thrombus in the aortic arch, which contraindicates endovascular navigation in the arch, and the anatomy of the supra-aortic trunks, in order to plan the radial or femoral approach and catheterization. In addition, the absence of an anatomical variant contraindicating the procedure will be verified: birth of the MMA from the ophthalmic artery, as evidenced by agenesis of the foramen spinosum and the absence of an MMA arising from the internal maxillary artery.

Embolization will be performed via a femoral or radial access. The middle meningeal artery will be catheterized under radiological control using a Progreat microcatheter, then embolized with calibrated particles to devascularize the downstream territory until truncal occlusion is achieved. This occlusion of an artery independent of cerebral vascularization has already been proposed in several indications, such as pre-operative embolization of hyper-vascular meningiomas, and has no known negative consequences in the absence of macroscopic anastomoses with cerebral arteries or the ophthalmic artery (24). This will be verified during the pre-intervention CT angiography. Indeed, the CT angiography allows us to search for the birth of the ophthalmic artery from the middle meningeal artery, and conversely, the birth of the middle meningeal artery from the ophthalmic artery.

If an anastomosis between the MMA and the ophthalmic artery is detected during catheterization and selective angiography of the MMA, embolization can be performed either by injecting the particles beyond the anastomosis with a sufficient safety margin, or after occlusion of the anastomosis by coiling, enabling protection of the ophthalmic artery before injection of the particles. The two main branches of the MMA (anterior-frontal and posterior-parietal or petrosquamous) can be embolized together via the common trunk of the MMA, or selectively sequentially to protect the orbital branches (from the anterior branch) or a prominent petrous branch.

Strict bed rest for 12 to 24 hours after use of the femoral approach will minimize the risk of hematoma at the puncture site.

It is important to note that it is the "embolization" procedure that we are evaluating. This procedure involves a medical device: CE-marked calibrated particles for vascular embolization, e.g. for hyper-vascular meningiomas.

For patients operated for bilateral cSDH and presenting a risk factor for recurrence such as antiplatelet therapy, if randomized to the intervention arm, bilateral embolization will be performed. A patient operated bilaterally for bilateral cSDH and not presenting risk factors for recurrence as set out above will not be eligible after the first surgery. In the event of recurrence leading to repeat surgery, whether unilateral or bilateral, they may be included in the absence of non-inclusion criteria. If randomized in the embolization arm, both sides, right and left, must be embolized (even if the recurrence surgery involved only one of the two sides, right or left).

2.6 Summary of foreseeable and known benefits and risks for research subjects

Demonstrating the efficacy of this procedure would have a major impact on the management of cSDH, with a reduction in the risk of post-operative recurrence, and consequently in the morbidity and mortality of this pathology. It would also reduce the rate of repeat surgery and length of hospital stay. The procedure would allow the resumption of anti-coagulants and antiplatelet medication for primary or secondary prevention in cases of formal indication.

This procedure would expose the patient to low risk (risk associated with endovascular navigation, exposure to X-rays and injection of a contrast medium). The procedure would require strict bed rest for 12 to 24 hours, and should not prolong hospitalization.

As the majority of patients with cSDH are elderly and vulnerable, the minimally invasive, low-risk nature of the proposed procedure seems particularly appropriate. The minimally invasive nature of the procedure also means that the post-intervention after-effects are very short, avoiding prolonged hospitalization.

3 OBJECTIVES

3.1 Main objective

To evaluate the efficacy of MMA embolization in reducing the risk of 6-month post-operative recurrence of cSDH surgically evacuated by burr-hole or trephine compared with usual management in patients at high risk of recurrence.

3.2 Secondary objectives

- 1/ Determine the impact of MMA embolization on the rate of disability and dependence at 1 month and 6 months compared with usual management.
- 2/ Determine the impact of MMA embolization on 1-month and 6-month mortality rates compared with usual management.
- 3/ Determine the impact of MMA embolization on cumulative length of hospital stay, in relation to cSDH, compared with usual management.
- 4/ Determine the impact of MMA embolization on the rate of re-intervention for homolateral recurrence of cSDH
- 5/ Determine the safety and tolerance of the procedure.

3.3 Objectives of potential ancillary studies

- 1/ Identify the clinical and radiological risk factors most significantly associated with postoperative recurrence of cSDH (in the control group).
- 2/ Describe the angiographic findings in patients with cSDH.
- 3/ Describe the angiographic anatomical variants of the middle meningeal artery.

4 RESEARCH DESIGN

4.1 Research evaluation criteria

4.1.1 Primary endpoint

CSDH recurrence rate, at 6 months, defined as :

- Reappearance of a homolateral cSDH with a midline shift ≥ 5 mm or a symptomatic homolateral SDH, including leading to death
- Or, the presence of a homolateral cSDH > 10 mm in maximal thickness on the 6 months control head CT scan

- Or, the need for repeated surgery for a homolateral cSDH recurrence
- Or, the need for a new hospital admission in relation to a homolateral SDH recurrence

For patients operated for bilateral cSDH, the primary endpoint is considered to have been met for any recurrence, whether unilateral or bilateral.

The primary endpoint will be assessed at a 6-month consultation with an interventional neuroradiologist following a follow-up brain scan. Recurrences occurring during the first month will be collected at the 1-month consultation. If an event occurs during follow-up, all hospitalization, consultation or imaging reports will be collected.

An independent, blinded adjudication committee will evaluate any recurrences on the basis of anonymized medical data submitted to the committee (imaging, consultation, surgery and hospitalization reports).

4.1.2 Secondary endpoints

- 1/ Proportion of dependent patients (modified Rankin score greater than or equal to 4) at 1 and 6 months
- 2/ Mortality rates at 1 month and 6 months
- 3/ Cumulative duration of hospitalization at 6 months, related to the initial cSDH, a recurrence or its direct or indirect consequences, including day hospitalization. Hospitalizations strictly independent of the cSDH or its consequences will not be counted.
- 4/ Rate of neurosurgical re-intervention for recurrence of homolateral cSDH during the study period. For patients operated on for bilateral cSDH, this secondary endpoint is considered met for any recurrence leading to surgery, whether unilateral or bilateral.
- 5/ Complication rates :

Major complications :

- Major complications at the puncture site (retroperitoneal hematoma or arterial pseudoaneurysm requiring surgical treatment)
- Degradation of renal function with a persistent drop in creatinine clearance of more than 10 points
- Neurological accident with permanent sequelae

- Others

Minor complications :

- Superficial hematoma at puncture site
- Transient deterioration of renal function
- Transient neurological attack or asymptomatic arterial dissection
- Others

Secondary endpoints will be assessed at a neurosurgical consultation at 1 month (part of the usual management of operated cSDH patients, with a particular focus on procedural complications), and at 6 months with an interventional neuroradiologist following a follow-up brain CT scan.

4.2 Description of research methodology

4.2.1 Experimental design

Multicenter, open-label, randomized controlled trial with two parallel groups in a 1:1 ratio to test the superiority of middle meningeal artery embolization over standard management for the prevention of postoperative recurrence of cSDH.

4.3 Description of measures taken to reduce and avoid bias

4.3.1 Subject identification

For the purposes of this research, subjects will be identified as follows:

center no. (3 numeric positions) - person selection order no. in center (4 numeric positions) - initial surname - initial first name

This reference is unique and will be kept for the duration of the research.

A randomization number will be assigned at the time of randomization. It will be in the following format: RXXX in accordance with the sponsor's procedures.

4.3.2 Randomization

After signing an informed consent form, eligible patients will be randomized 1:1 between embolization (experimental group) and usual management (control group) within 7 days of evacuation of the cSDH. Randomization will be carried out with small permuted blocks of random size and will be stratified according to center, whether the patient is being treated with an anticoagulant or an antiplatelet medication (yes or no), and whether the cSDH is unilateral or bilateral. It will be centralized via the e-crf.

5 PRACTICAL RESEARCH UNFOALDING

Before any examination or act related to the research, the investigator obtains the free, informed and written consent of the person undergoing the research, or of his or her legal representative where applicable.

Persons likely to undergo research mentioned in 1° of article L. 1121-1 of the French Public Health Code are entitled to a preliminary medical examination appropriate to the research.

5.1 Selection visit

The screening visit takes place during the initial post-operative hospitalization, immediately after surgical evacuation of the cSDH. It will be carried out by a neurosurgeon or interventional neuroradiologist. Inclusion and non-inclusion criteria will be verified.

In particular, the following will be checked:

- Patient's age
- Past medical history
- chronic alcoholism or liver disease
- allergy to iodinated contrast media (not to be confused with allergy to Betadine or seafood)
- Usual treatments, in particular long-term antiplatelet and anticoagulation medication. The indication for resumption of these treatments after surgery will be evaluated.

- Patient's state of autonomy before cSDH by modified Rankin score (mRS)
- Renal function by measuring creatinine levels, usually available in the medical record
- The possibility of pregnancy for women of childbearing age, if necessary by a beta-HCG assay

For patients fulfilling all inclusion criteria and none of the non-inclusion criteria, the patient and/or a family member or the trusted support person designated by the patient will be informed of:

- The risk of recurrence of the operated cSDH.
- Potential consequences of cSDH recurrence.
- The possibility of performing MMA embolization, with calibrated particles, under local anesthesia or conscious sedation, to prevent recurrence.
- Modalities and risks of embolization.
- The possibility of participating in the EMPROTECT randomized trial.

5.2 Inclusion visit (T0)

After a reflection period of no more than 24 hours following the screening visit, an inclusion visit will be held to obtain the patient's signed informed consent. If consent is signed, randomization will be carried out during the inclusion visit.

The inclusion visit can take place at the same time as the selection visit for patients who do not wish a reflection period.

Persons whose consent is sought	Who informs and obtains the person's consent	When is the person informed?	When is the person's consent obtained?
<i>the research subject or a family member/trusted third party</i>	<i>The investigator (interventional neuroradiologist or neurosurgeon)</i>	<i>During the screening visit, during hospitalization;</i>	<i>During the inclusion visit (T0)</i>

5.3 Research follow-up visit at one month (T1)

A follow-up visit is scheduled at one month (T1). It can take place no earlier than 7 days before the theoretical T1 date and no later than 15 days after this theoretical date. This visit is part of the usual follow-up of patients operated for cSDH.

The following data will be collected in the e-crf :

- The occurrence of an adverse event related or unrelated to embolization
- The occurrence of a recurrence of CSDH defined by :
 - recurrence of a cSDH with midline shift greater than or equal to 5mm or symptomatic,
 - or the need for neurosurgical re-intervention due to recurrence of homolateral cSDH,
 - or the need for re-hospitalization for a recurrence of homolateral cSDH
- Modified Rankin Score (mRS)
- Cumulative length of hospitalization since randomization

5.4 End of research visit at 6 months (T2)

An end-of-research consultation will be scheduled 6 months after inclusion, with a follow-up brain CT scan. This consultation can be carried out no earlier than 28 days before the calculated theoretical date of T2 and no later than 28 days after the theoretical date of T2.

The following data will be collected in the e-crf :

- The occurrence of an adverse event related or unrelated to embolization
- The occurrence of a recurrence of the cSDH defined by :
 - recurrence of cSDH with midline shift greater than or equal to 5mm or symptomatic (during the 6-month follow-up period after randomization or on the 6-month follow-up CT scan),
 - or the presence of a cSDH of more than 10mm maximum thickness on the CT scan performed at 6 months,
 - or the need for neurosurgical re-intervention for recurrence of homolateral cSDH within 6 months of randomization,
 - or the need for re-hospitalization for recurrence of homolateral CSDH within 6 months of randomization,
- Modified Rankin Score (mRS)
- Cumulative duration of hospitalization since randomization (start and end dates of hospitalization will be noted).

If this consultation does not take place, the patient, a legal representative, a relative or a caregiver will be contacted by telephone for a follow-up interview.

5.5 Expected duration of participation and description of the chronology and duration of the research.

Each patient will participate for 6 months.

Length of inclusion period	24 months
Duration of subject participation, including :	
• Maximum duration of the selection period:	7 days after surgery
• Duration of use of the medical device:	NA
• Follow-up period :	6 months
Total duration of research :	30 months

5.6 Summary table or diagram of research chronology

<i>Actions</i>	<i>D0-D7 after surgery: Selection visit</i>	<i>Within 24 hours of the Selection Visit: Inclusion Visit (T0)</i>	<i>D0-D7 after surgery: Embolization (intervention group)</i>	<i>M1 Follow-up visit (from T1- 7 days to T1+15 days)</i>	<i>M6 Follow-up visit (T2 +/- 28 days)</i>
<i>Checking eligibility criteria</i>	X				
<i>Information</i>	X				
<i>Written consent</i>		X			
<i>Medical history</i>	X				
<i>Clinical examination*</i>	X	X		X	X
<i>CT angiography of the supra-aortic trunks and circle of Willis (intervention arm)</i>			X (before embolization)		

<i>Brain CT scan</i>				<i>X</i>	<i>X</i>
<i>Follow-up consultation</i>				<i>X</i>	<i>X</i>
<i>Adverse events</i>			<i>X</i>	<i>X</i>	<i>X</i>

** Each act should have its own line, where possible and legible.*

5.7 Distinction between care and research

TABLE: Distinction between "care" and "research" procedures

Acts, procedures and treatments performed for research purposes	Acts, procedures and treatments related to <u>care</u>	Acts, procedures and treatments added by <u>research</u>
CT angiography of the supra aortic trunks and circle of Willis		Only in the intervention arm
MMA embolization		Only in the intervention arm
Consultations	M1 and M6	
Brain CT scan	M1 and M6	

6 ELIGIBILITY CRITERIA

6.1 Inclusion criteria

Patients

- aged 18 or over
- Operated for a cSDH recurrence or operated for a first episode of cSDH if at least one of the following recurrence risk factors is present:
 - Chronic alcoholism : daily alcohol consumption > 30g/day
 - Liver cirrhosis
 - Antiplatelet therapy or effective anticoagulant therapy with a formal indication to resume treatment after surgery
 - Thrombocytopenia (platelet count < 100 G/L)
 - cSDH surgery without use of external drain
- affiliated to a social security scheme (beneficiary or beneficiary's beneficiary) (excluding Aide Médicale d'État)
- signed informed consent or written informed consent obtained from family/trusted support person

6.2 Non-inclusion criteria

Patients :

- operated by craniotomy or twist-drill craniostomy

- beyond 7 days after initial cSDH surgery or cSDH recurrence surgery,
- dependent before the occurrence of cSDH with a modified Rankin score ≥ 4 ,
- with an estimated life expectancy of less than 6 months,
- with chronic renal failure (clearance less than 30 ml/min),
- during pregnancy,
- with a history of allergy to iodinated contrast media,
- Contraindication to conscious sedation AND local anesthesia (e.g. agitation making it impossible to perform the procedure under local anesthesia or conscious sedation),
- refusing to participate
- whose follow-up is deemed difficult (patients living abroad, socially isolated or in very precarious situations),
- under guardianship or curatorship.

6.3 Recruitment procedures

Justification of recruitment capacity with regard to the number of subjects to be included ;

Patients will be recruited during hospitalization for cSDH surgery following hematoma evacuation.

All participating centers have neurosurgery teams that routinely manage cSDHs. Participating INR teams have expertise in the endovascular treatment of complex cerebrovascular pathologies. Almost all the centers participating in the PDSES (Permanence des soins en établissements de santé) for neurosurgery/INR/neuro-resuscitation in the Ile-de-France region, and thus managing neurosurgical emergencies, have agreed to participate. This means that the protocol covers almost the entire population of the Ile de France region. In addition, the participation of several provincial teams will further increase the speed of inclusion and hence the feasibility of the study. The participating centers totaled more than 1,000 cSDH surgeries in 2017. Assuming an eligibility rate of 1/3 and an inclusion rate of 50% of eligible patients, the expected speed of inclusion is around 150 patients/year, giving an approximate recruitment time of 24 months.

	<i>Number of subjects</i>
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<i>Total number of subjects selected</i>	<i>342</i>
<i>Number of centers planned</i>	<i>12</i>
<i>Inclusion period (months)</i>	<i>24</i>
<i>Number of subjects / center / year</i>	<i>14</i>
<i>Number of subjects / center / month</i>	<i>1.2 on average</i>

6.4 Stopping rules

6.4.1. Criteria and procedures for premature discontinuation of procedures performed in connection with research or administration of the product

Different situations exist.

- In the event of randomization to the intervention arm, CT angiography of the supra-aortic trunks and circle of Willis will be performed systematically. On the basis of this examination, a small number of patients will be excluded from the embolization procedure in cases of floating thrombus in the aortic arch or birth of the MMA from the ophthalmic artery with agenesis of the foramen spinosum. These patients will remain in the study, will be followed up for up to 6 months and will also be included in the analysis according to the intention-to-treat principle.
- Temporary stop: the investigator must document the reason for the stop and its resumption in the subject's source file and CRF.
- Premature termination, but subject remains in research until end of participation, investigator must document reason for termination
- Premature termination and cessation of participation in research: the investigator must document the reason(s).

6.4.2. Criteria and procedures for premature termination of a subject's participation in research:

- Any subject may stop participating in research at any time, for any reason.
- The investigator may temporarily or permanently discontinue a subject's participation in the research for any reason that has an impact on the subject's safety or that would be in the subject's best interest.

- Subject lost to follow-up: the subject's outcome is unknown. The investigator must make every effort to recontact the subject, his/her relatives or his/her attending general practitioner (and document this in the source file) in order to know at least whether the subject is alive or dead, and if possible whether he has been hospitalized for a recurrence of his/her cSDH.
- In the event of premature termination of a subject's research, or withdrawal of consent, data concerning the subject collected prior to premature termination may be used.
- If the patient refuses to attend the follow-up visits, the study endpoints can be collected by telephone interview with the patient, his/her family and his/her general practitioner. This will be notified.
- In the event of serious adverse events, please refer to the corresponding vigilance paragraph.
- The eCRF should list the various reasons for discontinuing participation in the research:
 - Subject's personal reasons
 - Explicit withdrawal of consent.
 - Lost to follow-up

Procedures for replacing these people, if applicable: Patients lost to follow-up or withdrawing consent will not be replaced. A rate of 20% of patients lost to follow-up is included in the calculation of the number of subjects required.

6.4.3. Stopping all or part of the research project

The AP-HP sponsor or the Competent Authority (ANSM: agence nationale de sécurité des médicaments et des produits de santé; French national agency for security of medication and health products) may prematurely interrupt temporarily or definitively all or part of the research, following the recommendations of a Data Safety Monitoring Board (DSMB) in the following situations:

first, in the event of serious unexpected adverse events (SUSARS) in one arm of the study, or an imbalance of serious adverse events between the 2 arms of the study, necessitating a reassessment of the benefit/risk ratio of the study.

In case of significant difference ($p < 0.001$) in the planned interim analysis after inclusion of 129 subjects

Similarly, unforeseen events or new information relating to the procedure performed or the product used, in the light of which the objectives of the research or clinical program are unlikely

to be achieved, may lead the AP-HP sponsor or the Competent Authority (ANSM) to interrupt the research prematurely.

The AP-HP sponsor reserves the right to definitively suspend inclusions at any time if inclusion targets are not being met.

In the event of research termination, all patients already included will be followed up until the end-of-participation visit.

In the event of premature termination of research for safety reasons, the decision and justification are sent by the AP-HP sponsor within 15 days to the Competent Authority (ANSM) and to the ethical review board, together with the recommendations of the DSMB (if applicable) in the case of a substantial modification.

7 INTERVENTION AND MEDICAL DEVICES USED IN RESEARCH

7.1 MMA embolization procedure

For patients randomized in the experimental arm who have no contraindication to embolization based on data from the CT angiography of the supra-aortic trunks and circle of Willis, embolization will be performed as follows:

Under conditions of surgical asepsis, with local anesthesia of the puncture site and conscious sedation if required, via the femoral or radial artery if necessary, the external carotid artery homolateral to the hematoma is catheterized with a guide catheter. The MMA is catheterized beyond the foramen spinosum with a Progreat microcatheter navigated on a microguide wire. Hyperselective angiography of the middle meningeal artery through the microcatheter is performed to ensure that there is no macroscopic anastomosis between the MMA and the ophthalmic artery. In the absence of an anastomosis deemed risky by the operator, the MMA is embolized with calibrated particles, in free flow, until truncular occlusion.

If an anastomosis between the MMA and the ophthalmic artery is detected during catheterization and selective angiography, embolization can be performed either by injecting the particles beyond the anastomosis with a sufficient safety margin, or after occlusion of the

anastomosis by coiling, enabling protection of the ophthalmic artery prior to particle injection. The two main branches of the MMA (anterior-frontal and posterior-parietal or petrosquamous) can be embolized jointly via the common trunk of the MMA, or selectively sequentially to protect the orbital branches (from the anterior branch) or a prominent petrous branch.

The introducer sheath should be removed at the end of the procedure with compression or arterial closure device applied. Strict bed rest for 12 to 24 hours is recommended for femoral procedures, to limit the risk of hematoma at the puncture site.

It is the MMA embolization procedure that is the subject of this randomized trial. However, embolization implies the use of an implantable medical device (the embolization particles). As Merit Medical will be supplying the particles for the embolization procedures, the EmboSphere 300-500µm particles to be used are described in greater details in the following paragraphs.

For patients operated for bilateral cSDH and presenting a risk factor for recurrence such as antiplatelet therapy, if randomized to the intervention arm, bilateral embolization will be performed. A patient operated bilaterally for bilateral cSDH and not presenting risk factors for recurrence as set out above will not be eligible after the first surgery. In the event of recurrence leading to repeat surgery, whether unilateral or bilateral, they may be included in the absence of non-inclusion criteria. If randomized in the embolization arm, both right and left sides must be embolized (even if the recurrence surgery involved only one of the two sides, right or left).

7.2 Experimental device used for the MMA embolization procedure

7.2.1 Device identification

EMBOSPHERE 300-500µm calibrated embolization particles (Merit Medical)

7.2.2 Intended use of the device

The particles will be used to occlude the middle meningeal artery (embolization procedure) and delivered via a compatible microcatheter.

7.2.3 Summary of training and experience required to use the device

Any trained interventional neuroradiologist is able to perform a middle meningeal artery embolization procedure with calibrated particles without specific training. No additional technical training is required before performing this procedure, since all steps are routinely performed in other indications: 1) Femoral or radial arterial puncture 2) Catheterization of the external carotid artery 3) Microcatheterization of the middle meningeal artery 4) Calibrated particle embolization.

7.3 Comparator group description

Usual care including :

- Nursing care of the neurosurgical operating site
- Nursing care
- Symptomatic treatments: analgesics, for example
- Corticosteroid therapy if deemed necessary by the neurosurgical team
- Systematic clinical monitoring and imaging if necessary
- Rehabilitation if necessary

7.4 Description of embolization particles

EMBOSPHERE particles are non-absorbable, calibrated, biocompatible, hydrophilic particles impregnated with porcine gelatin. They are a derivative of acrylic polymer and porcine gelatin suspended in sodium chloride. The particles are packaged in syringes and vials.

7.5 Authorized and prohibited treatments (medicinal, non-medicinal, surgical), including rescue treatments

Heparin anticoagulation during the embolization procedure will be administered at the discretion of the interventional neuroradiologist. The procedure can be performed without heparin therapy.

For patients included because of a formal indication for resumption of effective anticoagulation or antiplatelet therapy, the time frame for resumption of treatment will be discussed between the referring neurosurgeon, interventional neuroradiologist, intensive care specialist, attending physician or drug prescriber. No time limit will be imposed by the study.

The administration of corticosteroids after cSDH surgery will be decided on a case-by-case basis by the patient's referring neurosurgeon.

All these elements will be monitored in the eCRF.

8 EVALUATION OF PERFORMANCE OR EFFECTIVENESS

8.1 Description of s performance or efficiency evaluation parameters

The primary endpoint is the rate of recurrence of cSDH at 6 months, defined as recurrence of cSDH with midline shift greater than or equal to 5mm or symptomatic, including leading to death, or the presence of a cSDH of more than 10mm maximum thickness on the 6-month follow-up CT scan, or the need for neurosurgical re-intervention for recurrence of homolateral cSDH during the study period, or the need for re-hospitalization for recurrence of homolateral cSDH.

Secondary performance or efficacy endpoints are the proportion of dependent patients (modified Rankin score greater than or equal to 4) at 1 and 6 months, mortality at 1 and 6 months, cumulative length of hospital stay at 6 months directly or indirectly related to cSDH, and the rate of re-intervention for a recurrence of cSDH during the 6-month period.

8.2 Methods and timetable for measuring, collecting and analyzing a performance or efficiency evaluation parameters

Primary and secondary endpoints will be assessed at a neurosurgery consultation at 1 month, and at an interventional neuroradiology consultation at 6 months, following a follow-up brain CT scan. If an event occurred during the follow-up period, all hospitalization, consultation and imaging reports will be collected.

A blinded evaluation of all potential recurrences according to the definition of the primary endpoint will be carried out by an independent adjudication committee. The medical record data required for this decision (imaging, consultation, hospitalization and intervention reports)

will be transmitted anonymously to the committee after removal of any information that could reveal the arm in which the patient was included.

Cases will be forwarded to the adjudication committee in the event of :

- Recurrence as defined in the primary endpoint, according to the local investigator
- Hospitalization, whatever the reason
- Consultation outside M1 and M6 scheduled consultations
- Follow-up cerebral imaging in addition to the 6-month follow-up CT scan
- New surgery of any kind.

9 SPECIFIC RESEARCH COMMITTEES

9.1 Steering Committee

9.1.1 Committee members :

Dr Eimad Shotar (coordinating investigator), Pr Frédéric Clarençon (scientific lead), Dr Agnes Dechartres (methodologist) as well as Mrs Anne Bissery (URC project manager) and a DRCI project manager (representatives of the sponsor and the URC).

9.1.2 Missions :

Define the general organization of the research, coordinate information, initially determine the methodology and monitor the research process.

Suggest actions to be taken during the course of the research, taking into account the recommendations of the DSMB, where applicable. The DRCI sponsor remains the decision-maker.

9.1.3 Practical organization :

Meetings scheduled before the start of the study and every 6 months for the duration of the study

9.2 Adjudication committee

9.2.1 Members:

The adjudication committee will consist of 3 members, neurosurgeon(s) and interventional neuroradiologist(s), all practicing in centers unrelated to the study.

9.2.2 Mission :

Blinded assessment of whether patients recurred as defined in the primary endpoint.

10 SAFETY ASSESSMENT - RISKS AND CONSTRAINTS ADDED BY RESEARCH

10.1 Definitions

According to article R.1123-46 of the French Public Health Code :

- **Adverse event (AE)**

Any harmful event occurring in a person undergoing research involving the human body, whether or not the event is related to the research or the product to which the research relates.

- **Adverse reaction to a medical device (MD) or *in vitro* diagnostic medical device (IVDD)**

Any harmful and undesired reaction to a medical device, or any incident that could have led to such a reaction if appropriate action had not been taken, in a person undergoing research or in the user of the medical device, or any effect related to a failure or alteration of an *in vitro* diagnostic medical device and harmful to the health of a person undergoing research.

- **Serious adverse event or reaction**

Any adverse event or effect that results in death, endangers the life of the person undergoing the research, requires hospitalization or prolongation of hospitalization, causes significant or lasting disability or incapacity, or results in a congenital anomaly or malformation, regardless of the dose administered for a drug.

- **Unexpected adverse reaction**

Any adverse reaction whose nature, severity or course is not consistent with the information on the products, procedures, practices and methods used in the research.

According to article R.5212-15 of the French Public Health Code :

- **Incident**

- Harmful and unintended reaction that occurs when a medical device is used for its intended purpose;
- Harmful and unintended reaction resulting from the use of a medical device that does not comply with the manufacturer's instructions;
- Any malfunction or alteration in the characteristics or performance of a medical device;

- Any errors, omissions or inadequacies in the instruction manual, operating instructions or maintenance manual.

According to article R.1123-46 of the French Public Health Code and the notice to sponsors of clinical trials involving medical devices and *in vitro* diagnostic medical devices (ANSM):

- **New fact**

Any new data that could lead to a reassessment of the risk-benefit ratio of the research or the product under investigation, to changes in the use of the product, in the conduct of the research, or in the documents relating to the research, or to the suspension, interruption or modification of the research protocol or similar research. For trials involving the first administration or use of a health product in people with no medical condition: any serious adverse event.

Examples:

- Any clinically significant increase in the incidence of an expected serious adverse event;
- Suspected unexpected serious adverse events in participants who have completed the trial, which are reported by the investigator to the sponsor, as well as any follow-up reports;
- Any new fact concerning the conduct of the clinical trial or the use of the medical device, when this new fact is likely to affect the safety of the participants.

For example:

- a serious adverse event likely to be linked to the trial's investigations and diagnostic procedures and which could alter the course of the trial,
- a significant risk for the trial population, such as a lack of efficacy of the medical device used in the treatment of a life-threatening disease,
- significant results from a completed preclinical study that may call into question the assessment of risk versus expected benefit (such as a biomechanics study),
- early termination or temporary interruption for safety reasons of a trial conducted with the same medical device in another country,
- an unexpected serious adverse event linked to a non-investigational health product required for the trial (e.g. challenge agents, rescue treatment),
- The recommendations of the DSMB, if any, if they are relevant to personal safety,
- Any unexpected serious adverse event transmitted to the sponsor by another sponsor of a clinical trial involving the same medical device in a third country.

10.2 The investigator's role

The investigator must **assess the seriousness of each adverse event**, and record all serious and non-serious adverse events in the e-CRF.

The investigator must **document** serious adverse events **to the best of his/her ability**, and give the definitive medical diagnosis whenever possible.

The investigator must **assess the intensity of** adverse events:

- In general terms:

Mild: tolerated by the patient, does not interfere with daily activities

Moderate: sufficiently uncomfortable to affect daily activities

Severe: prevents daily activities

The investigator must **assess the causal link** between serious adverse events and each medical device and/or the procedure used to install it and/or the specific acts/procedures/examinations added by the research, and with any other treatments.

The method used by the investigator, based on the WHO (WHO Uppsala Monitoring Centre) method, is based on the following 4 causal terms:

- Certain,
- Probable/plausible,
- Possible,
- Unlikely (not excluded).

Their definition is presented in the following table (extracted from WHO-UMC causality categories, version of 17/04/2012).

Table: WHO-UMC causality categories (extract)

Causality term	Assessment criteria*
Certain	· Event or laboratory test abnormality, with plausible time relationship to the study procedure · Cannot be explained by disease or other drugs · Response to withdrawal plausible (pharmacologically, pathologically) · Event definitive pharmacologically or phenomenologically (i.e. an objective and specific medical disorder or a recognized pharmacological phenomenon) · Rechallenge satisfactory, if necessary
Probable / Likely	· Event or laboratory test abnormality, with reasonable time relationship to the study procedure · Unlikely to be attributed to disease or other drugs · Response to withdrawal clinically reasonable · Rechallenge not required

Causality term	Assessment criteria*
Possible	· Event or laboratory test abnormality, with reasonable time relationship to the study procedure · Could also be explained by disease or other drugs · Information on drug withdrawal may be lacking or unclear
Unlikely	· Event or laboratory test abnormality, with a time to the study procedure · that makes a relationship improbable (but not impossible) · Disease or other drugs provide plausible explanations

*All points should be reasonably complied with

The investigator should specify whether the serious adverse event is the result of an incident related to the investigational medical device and/or its implementation procedure(s).

Any investigator who becomes aware of an incident must report it to the sponsor without delay. He/she fills in the incident or risk of incident reporting form (see appendices 19.3 and 19.4) and sends it by e-mail (eig-vigilance.drc@aphp.fr) to the DRCI Vigilance Sector (AP-HP), as soon as he/she becomes aware of it.

10.2.1 Serious adverse events requiring immediate notification by the investigator to the sponsor

The investigator must notify the sponsor without delay, from the day he/she becomes aware of them, of all serious adverse events occurring in the course of research mentioned in 1° of article L.1121-1, with the exception of those listed in the protocol (see corresponding section) and, where applicable, in the investigator's brochure as not requiring immediate notification (extract from article R.1123-49 of the French Public Health Code).

A serious adverse event meets one of the following criteria:

- 1- Event leading to death,
- 2- A life-threatening event for the person undergoing research,
- 3- Event requiring hospitalization or prolongation of hospitalization,
- 4- An event that causes significant or lasting disability or handicap,
- 5- An event that results in a congenital anomaly or malformation.

Serious adverse events that will be the subject of special monitoring by the sponsor include :

- any serious adverse event with a fatal outcome,
- fatal cSDH recurrences,

- neurological events,
- renal events,
- major complications at the puncture site,
- ophthalmological events,
- others.

10.2.2 Protocol specifics

10.2.2.1 Other events requiring immediate notification by the investigator to the sponsor

- **Adverse events deemed “medically significant”**

These adverse events must be notified to the sponsor by the investigator without delay from the day he/she becomes aware of them, according to the same procedures and deadlines as for serious adverse events (see above).

- ***In utero* exposure**

Any pregnancy occurring during the course of research, even if not associated with an adverse event, must be notified to the sponsor by the investigator without delay from the day on which he/she becomes aware of it.

10.2.2.1 Serious adverse events not requiring immediate notification by the investigator to the sponsor

These serious adverse events are only recorded in the eCRF. An extraction of these serious adverse events from the observation book will be made once a year, when the annual safety report is drawn up, and forwarded by the Clinical Research Unit to DRCI's Vigilance sector.

- **Natural and usual course of the disease:**

- Worsening of the pathology studied: recurrences of cSDH, with the exception of those with a fatal outcome, defined by :

- Recurrence of a cSDH greater than 10 mm or symptomatic during the study period

- Or, the need for neurosurgical re-intervention during the study period for recurrence of cSDH
- Or, the need for re-hospitalization related to the recurrence of homolateral cSDH during the study period.

- **Special circumstances**

*** Complications related to subdural hematoma and hematoma evacuation surgery:**

- Surgical site infection, subdural empyema
- Epileptic seizures occurring prior to the MMA embolization procedure
- Complications of general anesthesia
- Cerebral hemorrhage, pericerebral hemorrhage, acute subdural hematoma
- Inhalation pneumonitis

*** Other events :**

- Hospitalization for medical or surgical treatment scheduled prior to research.
- Admission for social or administrative reasons.

- **Adverse events likely to be related to treatments prescribed as part of care (e.g. corticosteroid therapy) during research follow-up.**

These adverse reactions must be reported by the investigator to the regional pharmacovigilance center.

- **Adverse events occurring after the investigational medical device has been put into service**

Adverse events occurring after the experimental medical device has been used will be reported without delay during the study period.

10.2.3 Period of immediate notification of SAEs by the investigator to the sponsor

The investigator must notify the sponsor without delay of serious adverse events as defined in the corresponding section:

- The date of the first act/procedure/examination specific to the research.

- For the duration of the participant's follow-up,
- Without time limit, when the SAE is likely to be due to the medical device and/or the acts/procedures/examinations specific to the research.

10.2.4 Procedures and deadlines for notifying the sponsor

Initial notification of a SAE is recorded in a written report signed by the investigator, using a research-specific SAE notification form (in the eCRF).

Each item in this document must be completed by the investigator to enable the sponsor to carry out a relevant analysis.

Initial notification of a serious adverse event to the sponsor must be followed up promptly by detailed written follow-up report(s) to monitor the progress of the case under vigilance or to supplement the information.

Whenever possible, the investigator will forward any documents that may be useful to the sponsor (medical reports, biological results, results of additional examinations, etc.). These documents must be made anonymous. In addition, they must be completed with the following information: research acronym, participant number and initials on all pages.

Any adverse event will be followed up until its complete resolution (stabilization at a level deemed acceptable by the investigator, or return to the previous state), even if the participant has withdrawn from the research.

Initial notification, SAE follow-up reports and any other documents will be sent to the sponsor, represented by the Vigilance Sector, exclusively by e-mail (eig-vigilance.drc@aphp.fr). Please note that the Vigilance Sector can be contacted by fax (01 44 84 17 99) only in the event of an unsuccessful attempt to send SAE reports by e-mail (to avoid duplication).

In the case of a search with e-CRF :

- The investigator completes the SAE notification form in e-CRF, validates it, prints it out, signs it and sends it by e-mail;

- if unable to connect to e-CRF, the investigator completes, signs and sends the SAE notification form to the Vigilance department. As soon as the connection is re-established, the investigator will fill it in the e-CRF form.

The investigator must respond to any request for additional information from the sponsor. For any questions about reporting an adverse event, it is possible to contact the Vigilance Sector by e-mail: vigilance.drc@aphp.fr.

In the event of *in utero* exposure, the investigator completes the "Pregnancy notification and follow-up form".

The investigator must follow the pregnant woman until the end of the pregnancy or its termination and notify the sponsor of the outcome with this form.

If the outcome of the pregnancy falls within the definition of serious adverse events (spontaneous abortion, termination of pregnancy, fetal death, congenital anomaly, etc.), the investigator must follow the procedures for reporting serious adverse events.

Initial pregnancy notification, SAE follow-up reports and any other documents will be sent to the sponsor in the same way as above.

10.3 The sponsor's role

The sponsor, represented by its Vigilance department, assesses the safety of each investigational medical device and/or its implementation procedure and/or the specific acts/procedures/examinations added by the research and with any other treatments, on an ongoing basis, throughout the research.

10.3.1 Analysis and reporting of serious adverse events

The sponsor evaluates :

- the **severity** of all adverse events reported,
- their **causal link** with each investigational medical device and/or its implementation procedure and/or specific acts/procedures/examinations added by the research and with any other treatments,
- all serious adverse events for which the investigator and/or sponsor consider that a causal relationship with the investigational medical device and/or its implementation procedure and/or

acts/procedures/examinations specific to the research and with other possible treatments can be reasonably envisaged are considered as suspected serious adverse events.

- the **expected or unexpected nature** of serious adverse reactions.

Any serious adverse reaction whose nature, severity or course are not consistent with the information given in the instruction leaflet or instructions for use when the medical device is CE-marked, or in the investigator's brochure when the product is not authorized, is considered unexpected.

The evaluation of the expected/unexpected nature of a serious adverse reaction is carried out by the sponsor, represented by its Vigilance department, on the basis of the information described below.

Serious adverse events likely to be related to the investigational medical device and/or its implementation procedure:

- refer to the instruction leaflet or instructions for use when the medical device is CE-marked: "EmboSphere® " leaflet included in appendix 19.6;

*Complications associated with the middle meningeal artery embolization procedure :

- Puncture-site complications: superficial hematoma with deglobulation (loss of 2 Hb points) and/or requiring transfusion, retroperitoneal hematoma with or without deglobulation, arterial pseudoaneurysm at puncture site requiring surgical treatment, femoral artery occlusion and/or acute limb ischemia, puncture-site abscess.
- Degradation of renal function with a persistent drop in creatinine clearance of more than 10 points
- Severe allergic reaction to iodinated contrast media: anaphylactic shock, angioedema, bronchospasm with desaturation, widespread skin rash.
- Severe allergic reaction to embolization product
- Asymptomatic occlusive or stenosing dissection of a cervico-encephalic artery
- Symptomatic dissection of a cervico-encephalic artery or one requiring salvage stenting
- Symptomatic intracranial cerebral emboli or those requiring mechanical thrombectomy
- Migration of embolic agent into the ophthalmic artery with visual acuity loss or visual field amputation
- Post-radiation complications, in particular transient post-radiation alopecia or permanent alopecia

- Serious complications of anesthesia in the event of sedation
 - Inhalation pneumonitis
 - Neurological accident with permanent sequelae
 - Severe infection
- Please also refer to the current summary of characteristics of the contrast medium used.
- Please also refer to the current summary of product characteristics for heparin therapy.
- Please also refer to the current summary of product characteristics for conscious sedation.
- Please also refer to the current summary of product characteristics for local anaesthesia.
- Exposure to ionizing radiation can induce cancer or the development of hereditary defects.
- Given the doses used for this examination, the occurrence of these adverse effects is unlikely.

Serious adverse events likely to be linked to specific research procedures:

*Complications associated with CT angiography of the circle of Willis:

- Degradation of renal function with a persistent drop in creatinine clearance of more than 10 points
 - Severe allergic reaction to iodinated contrast media: anaphylactic shock, angioedema, bronchospasm with desaturation, widespread skin rash.
 - Extravasation of contrast medium with skin necrosis
 - Decompensation of heart failure
- Urinary tract infection complicated by pyelonephritis
- Pulmonary embolism
- Pressure sores
- Please also refer to the current summary of characteristics of the contrast medium used.
- Exposure to ionizing radiation can induce cancer or the development of hereditary defects.
- Given the doses used for this examination, the occurrence of these adverse effects is unlikely.

The sponsor declares :

- any suspected serious unexpected adverse reaction to an investigational medical device (SUSAR),

- all serious adverse events likely to be linked to the implementation of the investigational medical device being the object of the research,
- any incident leading to a serious adverse event.

NB: the sponsor declares any incident leading to the occurrence of a serious adverse event to the ANSM and the ANSM's materiovigilance unit.

The sponsor reports any suspected serious unexpected adverse reaction (SUSAR) or serious adverse event that may be related to the procedure for implementing the medical device occurring in France and outside the national territory concerned, within the regulatory deadlines, to the Agence nationale de sécurité des médicaments et des produits de santé (ANSM):

- The initial report must be made without delay from the day on which the sponsor becomes aware of it in the case of serious unexpected adverse reactions and serious adverse events which may be related to the procedure for implementing the medical device and which have resulted in death or are life-threatening, and within 15 days from the day on which the sponsor becomes aware of it in the case of other serious unexpected adverse reactions and serious adverse events which may be related to the procedure for implementing the medical device;
- In the event of a serious unexpected adverse reaction or serious adverse event that may be related to the procedure for implementing the medical device, resulting in death or is life-threatening, the relevant additional information is notified to the French National Agency for the Safety of Medicines and Health Products (ANSM) within eight days.
- In other cases of serious unexpected adverse reactions and serious adverse events that may be linked to the procedure used to implement the medical device, the relevant additional information is sent to the Agence nationale de sécurité du médicament et des produits de santé (ANSM) within a further fifteen days.

The sponsor informs all investigators concerned of any data that could have an adverse impact on the safety of research subjects.

10.3.2 Analysis and declaration of other safety data

According to article R.1123-46 of the French Public Health Code (Code de la Santé Publique) and the notice to sponsors of clinical trials involving medical devices and *in vitro* diagnostic

medical devices (ANSM), a new fact is defined as any new data that may lead to a reassessment of the benefit/risk ratio of the research or of the investigational product, to changes in the use of this product, in the conduct of the research, or in the documents relating to the research, or to the suspension, interruption or modification of the research protocol or similar research. For trials involving the first administration or use of a health product in people with no medical condition: any serious adverse reaction.

The sponsor informs the competent authority and the ethical review board (comité de protection des personnes) without delay of new facts and, where appropriate, of measures taken, as from the day on which he becomes aware of them.

Following the initial declaration of a new fact, the sponsor sends the competent authorities, in the form of a follow-up report on the new fact, any relevant additional information relating to this new fact within a maximum of 15 days from the time when this information becomes available.

10.3.3 Annual safety report

The sponsor must draw up an annual safety report (ASR) once a year for the duration of the clinical trial, including in particular :

- an analysis of the safety of research subjects,
- a list of all suspected serious adverse events and a list of serious adverse events that may be linked to the medical device implementation procedure, occurring during the period covered by the report,
- summary tables of all serious adverse events that have occurred since the start of the study.

The report is sent within 60 days of the anniversary date corresponding to the date of the first patient included in the research.

10.4 Data and Safety Monitoring Board (DSMB)

The DSMB can be set up by the sponsor. Its main mission is to monitor safety data. It may have additional missions, such as monitoring efficacy data (particularly if the protocol provides for interim analyses).

It is up to the sponsor to justify to the competent authority (ANSM) and to the ethical review board whether or not such a committee has been set up.

A DSMB is planned as part of this research. A preliminary meeting of the DSMB is scheduled before the first participant is included.

The DSMB will be composed by an independent interventional neuroradiologist, neurosurgeon and methodologist. The DSMB is composed of Pr Bruno GIRAUDEAU (methodologist), Dr Paolo MACHI (interventional neuroradiologist) and Dr Fabian BAUMANN (neurosurgeon).

All the DSMB's missions and precise operating procedures are described in the study's DSMB charter.

The DSMB has an advisory role, while the sponsor remains the decision-maker.

11 DATA MANAGEMENT

11.1 Data collection methods

Data at T0, T1 and T2 will be collected electronically (e-crf) by the investigator at each center. Data entry will be anonymous. The complementary work of the clinical study technicians and the URC clinical research assistant (CRA) will be to ensure the accuracy and completeness of data collection.

11.2 Identification of data collected directly in CRFs to be considered as source data

No

11.3 Right of access to source data and documents

11.3.1 Data access

In accordance with GCP (good clinical practice) :

- the sponsor is responsible for obtaining the agreement of all parties involved in the research to guarantee direct access to all research sites, source data, source documents and reports for the purposes of quality control and audit by the sponsor or inspection by the competent authority,
- investigators will make available to those responsible for monitoring, quality control, audit or inspection of research involving the human person, the documents and individual data strictly

necessary for such control, in accordance with the legislative and regulatory provisions in force (articles L.1121-3 and R.5121-13 of the French Public Health Code).

11.3.2 Source documents :

Source documents are defined as any original document or object that can be used to prove the existence or accuracy of data or facts recorded during the course of the research, and will be kept by the investigator, or by the hospital in the case of hospital medical records, in accordance with current regulations.

Source documents :

- Medical file
- Hospitalization reports, consultation reports and correspondence
- Intervention reports
- Imaging test reports
- Biological and anatomopathological test results
- Imaging data in DICOM format

11.3.3 Data confidentiality :

Those responsible for quality control in research involving the human body (article L.1121-3 of the French Public Health Code), will take all necessary precautions to ensure the confidentiality of information relating to the research, to the persons involved, and in particular to their identity, and to the results obtained.

These people, like the investigators themselves, are bound by professional secrecy (under the conditions defined by articles 226-13 and 226-14 of the French penal code).

During or at the end of research involving the human person, data collected on subjects and transmitted to the sponsor by the investigators (or any other specialist) will be rendered non-identifying.

Under no circumstances may the names or addresses of the persons concerned appear in clear text.

Only the initials of the surname and first name will be recorded, together with a coded number specific to the research indicating the order of inclusion of subjects.

The sponsor will ensure that each person taking part in the research has given his or her written consent for access to individual data concerning him or her and strictly necessary for the quality control of the research.

11.4 Data processing and storage of documents and data

11.4.1 Identification of the data controller and data processing location

Data management and statistical analysis will be carried out at the Pitié-Salpêtrière URC, under the responsibility of Dr Agnès Dechartres.

11.4.2 Data entry

Data will be entered electronically (e-CRF) via a web browser.

11.5 Ownership of data

The AP-HP is the owner of the data and no use or transmission to a third party may be made without its prior agreement.

12 STATISTICAL ASPECTS

12.1 Description of planned statistical methods, including schedule of planned interim analyses

A patient flow diagram will be drawn up to describe the number of eligible patients and the number of patients actually included and randomized in total and in each of the two arms, as well as patient follow-up (number of patients lost to follow-up per arm). This diagram will also show the number of patients analyzed in each arm.

For each group and each evaluation date, qualitative variables will be described by their number, percentage and missing data per response modality, and quantitative variables by their number, mean and standard deviation. Quantitative variables not normally distributed will be presented with their median and interquartile range (1st quartile-3^e quartile).

Analysis of primary endpoint

The primary endpoint will be analyzed sequentially in 2 stages: in an interim analysis planned at the end of the 6-month follow-up of 129 included patients (37.5%), approximately 15 months after the start of inclusions (9 months of inclusion + 6 months of follow-up), and at the end of the study (at the end of follow-up of the last included patient, data collection and control). This sequential analysis is based on a 2-stage Lan & DeMets approach with an O'Brien & Fleming alpha risk expenditure function.

For the interim analysis, all patients randomized and for whom recurrence status at 6 months could be recovered will be analyzed, in their initially allocated randomization group. Patients who died without recurrence will be considered failures. Recurrences of cSDH by group will be analyzed using a mixed logistic model adjusted for randomization stratification factors. Randomization group, anticoagulant/antiplatelet use and uni or bilateral cSDH will be included as fixed effects, and center will be introduced as a random effect. In the event of model non-convergence, a Mantel-Haensel test (adjusted for center and, if possible, other randomization stratification factors) will be used.

In the event of an early stopping decision following this interim analysis, due to rejection of the hypothesis of equality of cSDH recurrence rates between the two groups at the nominal threshold of $\alpha=0.001$, the final main analysis will include the patients analyzed leading to the stopping decision, as well as all patients for whom the primary endpoint could be collected subsequently (over-running patients). Similarly, included patients who had already undergone embolization will continue to be followed up until M6, and will be included in the final analysis.

During the interim analysis, conditional power estimates will also be made. If the conditional power is deemed to be too low, the project may be discontinued on grounds of futility, after consultation with the DSMB.

In the final primary analysis, the primary endpoint, cSDH recurrence rate, will be analyzed in the intention-to-treat population using a mixed logistic model adjusted for randomization stratification factors. Randomization group, anticoagulant/antiplatelet medication use and uni or bilateral cSDH distribution will be included as fixed effects, while center will be introduced as a random effect (to take into account the multicenter nature of the study).

In the event of model non-convergence, a Mantel-Haensel test (adjusted for center and, if possible, other randomization stratification factors) will be used. Patients who die without recurrence will be considered as failures.

The low significance level chosen for the interim analysis during the study leads to a virtual absence of alpha risk inflation; the final analysis will therefore be carried out at the conventional 0.05 level, despite the interim analysis.

A sensitivity analysis will be performed to exclude patients who did not benefit from embolization in the experimental arm after CT angiography.

Two subgroup analyses will be carried out according to the uni or bilateral nature of cSDH and the use of anticoagulants/antiplatelet medication.

Analysis of secondary endpoints

The proportion of dependent patients at 1 and 6 months (Rankin score ≥ 4) will be compared between the two arms of the study using a GEE logistic model, with a cluster effect on the

patient to account for intra-patient variability. This model will include as parameters the randomization arm, the visit as class and the visit*randomization arm interaction. The interaction terms will be used to determine whether there is a difference between the two study arms at one month, or 6 months.

Mortality at 1 and 6 months, cumulative hospital stay at 6 months, rate of re-intervention for recurrence of cSDH during the 6-month period, and complication rate will also be compared between the experimental group (MMA embolization) and the control group (standard care). Quantitative endpoints will be compared using Student's t-tests (or non-parametric Wilcoxon test in case of non-normal distribution). Qualitative criteria will be analyzed using the Chi-2 test (or Fisher's exact test if appropriate).

Adverse events observed at 1 and 6 months will be described globally and by treatment group, in terms of number, frequency and percentage of patients concerned. Serious adverse events will be described separately.

12.2 Assumptions for calculating the number of subjects required and results

To demonstrate a reduction in the recurrence rate from 15% (control group) to 5% (embolized group) with 80% power, and ensuring an overall bilateral alpha risk of 5%,

142 patients per group are required. This number was estimated by taking into account the 2 sequential tests planned according to the method of Lan and Demets with an O'Brien & Fleming alpha risk expenditure function (interim analysis after inclusion and end of follow-up of 129 patients (information fraction of 37.5%) at nominal threshold $\alpha=0.001$ and final analysis at threshold $\alpha=0.05$). Assuming a lost-to-follow-up rate of 20%, a total of 342 patients (171 per group) is required.

12.3 Expected statistical significance

All analyses will be performed at 5% two-sided alpha risk (with the exception of the interim analysis performed at the nominal threshold of 0.001 from 129 patients).

12.4 Statistical criteria for discontinuing research.

An interim analysis is planned after inclusion and end of follow-up of 129 patients for this study. Inclusion may be stopped after completion of this analysis in the event of a difference

between the two groups attested at the 0.001 threshold, or for reasons of futility if the conditional power is too low.

The research will end at the end of the follow-up of the last included patient.

12.5 Method for taking into account missing, unused or invalid data

The main analysis will be conducted on an intention-to-treat basis. All randomized subjects will be analyzed (in the randomization group initially allocated to them). In the main analysis, a multiple imputation method will be used for patients who cannot be evaluated for the primary endpoint.

12.6 Manage changes to the initial strategy analysis plan.

Any modification of the initial analysis plan will be justified in the study report.

12.7 Population selection

Intention to treat (ITT)

The ITT population is defined as all patients randomized to the study, with each patient remaining in the arm to which he or she was randomly assigned. The analysis of this population constitutes the main analysis of the study.

13 CONTROL AND quality assurance

Each interventional research project involving humans undertaken by the AP-HP is classified according to the anticipated risk incurred by the people taking part in the research, using the classification of interventional research involving humans promoted by the AP-HP.

13.1 General organization

The sponsor must ensure the safety and respect of those who have agreed to take part in the research. They must set up a quality assurance system to monitor the progress of the research in the investigating centers.

To this end, the sponsor appoints Clinical Research Assistants (CRAs), whose main task is to make regular follow-up visits to the research sites after the opening visits.

The objectives of research follow-up, as defined in Good Clinical Practice (GCP § 5.18.1), are to verify that :

- the rights, safety and protection of research subjects are satisfied,
- reported data are accurate, complete and consistent with source documents,
- the research is conducted in accordance with the protocol in force, with GCP and with the legislative and regulatory provisions in force.

13.1.1 Scope of center monitoring :

In the case of this research, the choice of a suitable monitoring level was weighted according to the complexity, impact and budget of the research.

To this end, the sponsor, in agreement with the coordinating investigator, determined the logistical and impact score which made it possible to obtain the level of monitoring to be implemented on the research: high level.

These different levels are defined in the charter for the monitoring of Research Involving the Human Person.

13.2 Quality control

A Clinical Research Assistant (CRA) appointed by the sponsor will ensure that the research is carried out properly, and that the data generated is documented, recorded and reported, in accordance with the Standard Operating Procedures implemented within DRCI and in compliance with Good Clinical Practice and current legislative and regulatory provisions.

The investigator and the members of his or her team agree to make themselves available for the quality control visits carried out at regular intervals by the Clinical Research Associate.

During these visits, the following elements will be reviewed:

- written consent ;
- compliance with the research protocol and the procedures defined therein;
- quality of data collected in the eCRF: accuracy, missing data, consistency of data with "source" documents (medical records, appointment books, original laboratory results, etc.);

- device management.

13.3 Observation notebook

The eCRF should contain only the data required for analysis with a view to publication. All data relating to the subject and necessary for his/her follow-up during and outside the research will be collated in his/her medical file.

Electronic CRF :

All information required by the protocol must be recorded in the CRF. Data should be collected as they are obtained, and recorded explicitly in these CRFs. Any missing data should be coded. This electronic case report form will be set up in each center using an Internet data collection medium. Investigators will be provided with a document to help them using this tool.

If the investigator fills in the case report form via the Internet, the CRA can view the data quickly and remotely. The investigator is responsible for the accuracy, quality and relevance of all data entered. In addition, when data is entered, it is immediately checked for consistency. As such, he/she must validate any value modification in the CRF. These modifications are subject to an audit trail. A justification may be included as a comment.

A paper printout will be requested at the end of the study, authenticated (dated and signed) by the investigator. The investigator must keep a copy of the authenticated document for the sponsor.

13.4 Non-conformity management

Any event arising from non-compliance with the protocol, standard operating procedures, good clinical practice or applicable laws and regulations by an investigator or any other person involved in the conduct of research must be reported as a non-conformity to the sponsor.

These non-conformities will be managed in accordance with the sponsor's procedures.

13.5 Audits / inspections

Investigators agree to accept quality assurance audits by the sponsor and inspections by competent authorities. All data, documents and reports are subject to regulatory audits and inspections without prejudice to medical confidentiality.

An audit may be carried out at any time by persons mandated by the sponsor and independent of those in charge of the research. Its purpose is to ensure the quality of the research, the validity of its results and compliance with the law and regulations in force.

The persons directing and supervising the research agree to comply with the requirements of the sponsor and the competent authority with regard to an audit or inspection of the research.

The audit can be applied to all stages of research, from protocol development to publication of results, and to the classification of data used or produced as part of the research.

13.6 Responsibilities of the Principal Investigator

Before starting the research, each investigator will provide the research sponsor's representative with an updated, dated and signed personal curriculum vitae, including his/her RPPS number. The CV should include previous research participation and training related to clinical research.

Each investigator undertakes to respect the obligations of the law and to conduct the research in accordance with G.P.C., respecting the terms of the Declaration of Helsinki in force.

The principal investigator at each participating center will sign an undertaking of responsibility (standard DRCI document), which will be given to the sponsor's representative.

The investigators and their collaborators will sign a delegation of functions form specifying each person's role.

14 ETHICAL AND LEGAL ASPECTS

14.1 Information and consent procedures for research subjects

Who is informed? The participant, or if applicable: family members/trusted third party.

Who gives consent? The patient, or a close family member/trusted support person if the patient is unable to give consent.

When is it done? At the screening visit, within 7 days of cSDH evacuation surgery, during the same hospitalization.

How is this done? An information note is given to the participant or a family member/trusted third party, and verbal information is provided.

By whom? The investigator

In accordance with article L. 1122-1-1 of the French Public Health Code, no interventional research involving the human body may be carried out on a person without his or her free and informed consent, obtained in writing after the person has been provided with the information specified in article L. 1122-1 of the same code.

A 24-hour reflection period is allowed between the time the person is informed and the time he or she signs the consent form.

The person's free, informed and written consent is obtained by the principal investigator, or by a physician representing him/her, before the person is included in the research. Consent is obtained during the same hospitalization.

The information note and a copy of the consent form, dated and signed by the person undergoing the research and by the investigator or the physician representing him or her, are given to the person prior to participation in the research.

In addition, the investigator will specify in the patient's medical record the person's participation in the research, the procedures for obtaining his/her consent or the consent of any other person in the cases provided for in articles L. 1122-1-1 to L. 1122-2 of the CSP, and the procedures for providing the information needed to obtain it. He/she keeps a copy of the consent form, dated and signed.

Special case: If it is physically impossible for the person to give written consent, consent is attested by a third party. This third party must be totally independent of the investigator and the sponsor.

Special case: Mention of the possibility for the investigator to reserve certain information related to the diagnosis, where applicable in accordance with paragraph 4 of article L1122-1.

14.2 Prohibition on participation in other research

During this inclusion period, the subject may not participate in any other interventional research protocol involving the human body.

Subjects may, however, participate in other non-interventional research projects

14.3 Subject compensation

No compensation is provided.

14.4 Registration in the national file of persons undergoing research involving the person

No registration is planned as none of the situations described below apply.

In accordance with article L. 1121-16 of the French Public Health Code, registration is required:

- for persons with no medical condition who voluntarily agree to undergo research;
- for patients when the purpose of the research is unrelated to their medical condition;
- in certain special cases at the request of the ethical review board ;

14.5 Site authorization

The research is carried out in health-care departments on people with a clinical condition for which the departments are competent, and requires procedures normally performed as part of their activities. It is therefore not necessary to obtain authorization for a specific research site.

15 LEGAL OBLIGATIONS

15.1 The sponsor's role

Assistance Publique Hôpitaux de Paris (AP-HP) is the sponsor of this research and, by delegation, the Délégation à la Recherche Clinique et à l'Innovation (DRCI) is in charge of its missions, in accordance with article L.1121-1 of the French Public Health Code. Assistance Publique - Hôpitaux de Paris reserves the right to interrupt the research at any time for medical or administrative reasons; in this eventuality, a notification will be provided to the investigator.

15.2 Submission to the ethical review board

For interventional research involving the human body and involving a medical device or an in vitro diagnostic medical device, the AP-HP, as sponsor, obtains a favorable approval from the relevant ethical review board prior to implementation, within the scope of its competence and in accordance with the legislative and regulatory provisions in force.

15.3 Application for ANSM authorization

For interventional research involving the human body and involving a medical device or an in vitro diagnostic medical device, the AP-HP, as sponsor, obtains prior authorization from the ANSM, within the scope of its competence and in accordance with the legislative and regulatory provisions in force.

15.4 Information technology and civil liberties

The computer file used for this research is implemented in accordance with French (amended Data Protection Act) and European (General Data Protection Regulation -RGPD).

- Commitment to comply with MR 001 "Reference Methodology"

This research falls within the scope of the "Reference Methodology for the Processing of Personal Data Implemented in the Context of Healthcare Research" (MR-001 modified). AP-HP, the research sponsor, has signed a commitment to comply with this "Reference Methodology".

15.5 Research modifications

Any substantial modification made to the protocol by the coordinating investigator must be transmitted to the sponsor for approval. Once approved, the sponsor must obtain a favorable approval from the ethical review board and authorization from ANSM, within their respective areas of competence.

The information note and consent form may be revised if necessary, particularly in the event of substantial changes to the research or the occurrence of adverse effects.

15.6 Final research report

The final report on interventional research involving the human body, mentioned in article R1123-60 of the French public health code, is drawn up and signed by the sponsor and the investigator. A summary of the report, drawn up in accordance with the competent authority's

reference plan, must be sent to the competent authority within one year of the end of the research, corresponding to the end of the participation of the last person involved in the research.

15.7 Archiving:

Documents specific to research involving the human body that does not concern a health product will be archived by the investigator and the sponsor for *to be completed depending on the type of data (the choice of duration depends on the type of research)* after the end of the research.

This indexed archive includes :

- A sealed envelope for the investigator containing a copy of all the information notes and consent forms signed by all the people at the center who took part in the research;
- A sealed envelope for the sponsor containing a copy of all the information notes and consent forms signed by all the people at the center who took part in the research;
- “Research” binders for the Investigator and Sponsor including (non-exhaustive list):
 - successive versions of the protocol (identified by version number and date), its appendices
 - ANSM authorizations and ethical review board approvals
 - correspondence,
 - inclusion list or register,
 - research-specific appendices
 - the final research report.
- Data collection documents.

16 FINANCING AND INSURANCE

16.1 Source of financing

PHRC-Interregional.

16.2 Insurance

The Sponsor, for the entire duration of the research, takes out insurance covering its own civil liability as well as that of any physician involved in carrying out the research. It also guarantees full compensation for the harmful consequences of the research for the person taking part and his or her beneficiaries, unless it can prove that the damage is not attributable to its own fault or to that of any other party involved, without being able to invoke the act of a third party or the voluntary withdrawal of the person who had initially agreed to take part in the research.

Assistance Publique- Hôpitaux de Paris (AP-HP) has taken out an insurance policy with HDI-GERLING through BIOMEDIC-INSURE for the entire duration of the research, covering its civil liability as well as that of all participants (physicians or personnel involved in the research), in accordance with article L.1121-10 of the French public health code.

17 PUBLICATION RULES

17.1 Mention of AP-HP affiliation for projects promoted by AP-HP

- If an author has several affiliations, the order in which the institutions are cited (AP-HP, University, INSERM...) is irrelevant.
- Each of these affiliations must be identified by an address separated by a semicolon (;).
- The institution AP-HP must appear under the acronym "AP-HP" first in the address followed precisely by : AP-HP, hospital, department, city, zip code, France

17.2 Mention of the AP-HP sponsor (DRCI) in the manuscript acknowledgments

"The sponsor was Assistance Publique - Hôpitaux de Paris (Délégation à la Recherche Clinique et à l'Innovation)"

17.3 Mention of the funder in the manuscript acknowledgments

Concerning a PHRC: "The study was funded by a grant from Programme Hospitalier de Recherche Clinique - PHRC 2018 (Ministry of Health)".

This search will be registered on <http://clinicaltrials.gov/> (*add the registration number when the search is registered*).

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19 LIST OF ADDENDA

19.1 List of Investigators

19.2 List of pairs for each center

19.3 Serious Adverse Event notification form

19.4 Incident or malfunction report form

19.5 Pregnancy notification and follow-up form for research involving the human body

19.6 CE marking, instructions for use (if CE-marked DM)

The following documents relating to EmboSphere® particles are attached:

- EmboSphere CE mark

19.7 Radiological classification of cSDH